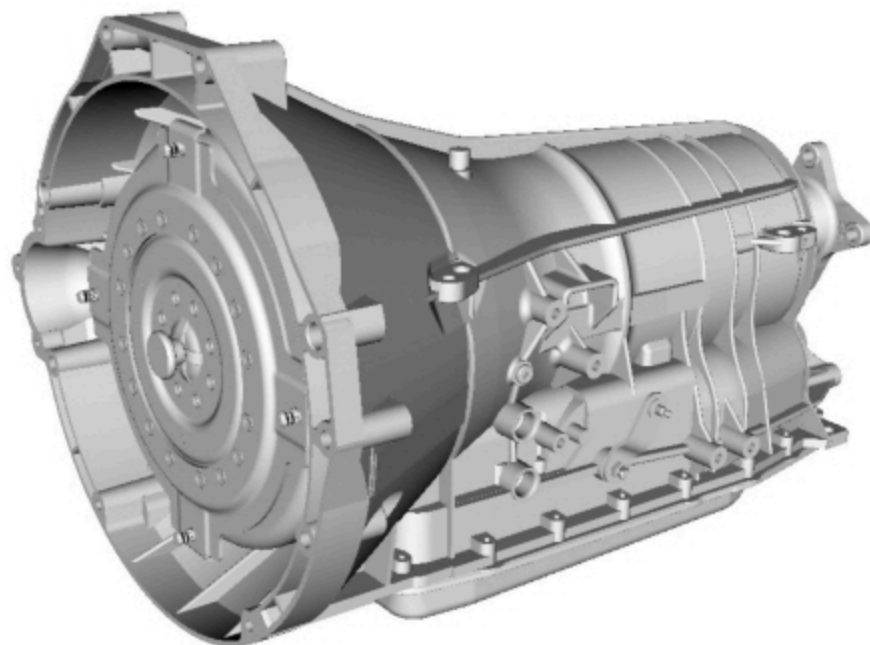


Transmission Description

 [Ford Ecat Electronic Parts Catalog](#)

6R80 Transmission



N0124989

This transmission has the following features:

- Six forward speeds
- Torque converter with an integral converter clutch
- Electronic shift and pressure controls
- Single planetary gear set
- Double planetary gear set
- Two fixed multi-disc clutches
- Three multi-plate clutches
- One-Way Clutch (OWC)

Hydraulic functions are directed by electronic solenoids to control the following:

- Engagement feel
- Shift feel
- Shift scheduling
- Modulated Torque Converter Clutch (TCC) applications

The 6R80 transmission has a main control assembly which contains:

- Turbine Shaft Speed (TSS) sensor
- Output Shaft Speed (OSS) sensor
- An internal Transmission Range (TR) sensor
- Transmission Fluid Temperature (TFT) sensor
- Line Pressure Control (LPC)

- TCC solenoid
- Five electronically controlled shift solenoids

Engine power reaches the transmission by a torque converter with an integral converter clutch. The 6 forward gears and one reverse gear are obtained from 2 planetary gearsets.

This 6-speed electronically controlled automatic transmission is comprised of a PCM, main control valve body assembly, torque converter, 1 solenoid valve and 6 pressure regulators. Gear selection is achieved by controlling transmission fluid to operate various internal clutches. The PCM operates the electrical components and provides for the control of gear selection and shift pressure, which increases refinement and torque converter slip.

In the event of a system fault, the PCM also provides for Failure Mode Effect Management to maintain maximum functional operation of the transmission with a minimum power reduction. In the event of a total loss of control or electrical power, the basic transmission functions park, reverse, neutral and drive are retained. Also 3rd or 5th gear is retained by the transmissions hydraulic system. The default gear retained is dependent upon the gear selected at the time of a failure.